

CSE 311:

Data Communication

Instructor:

Dr. Md. Monirul Islam

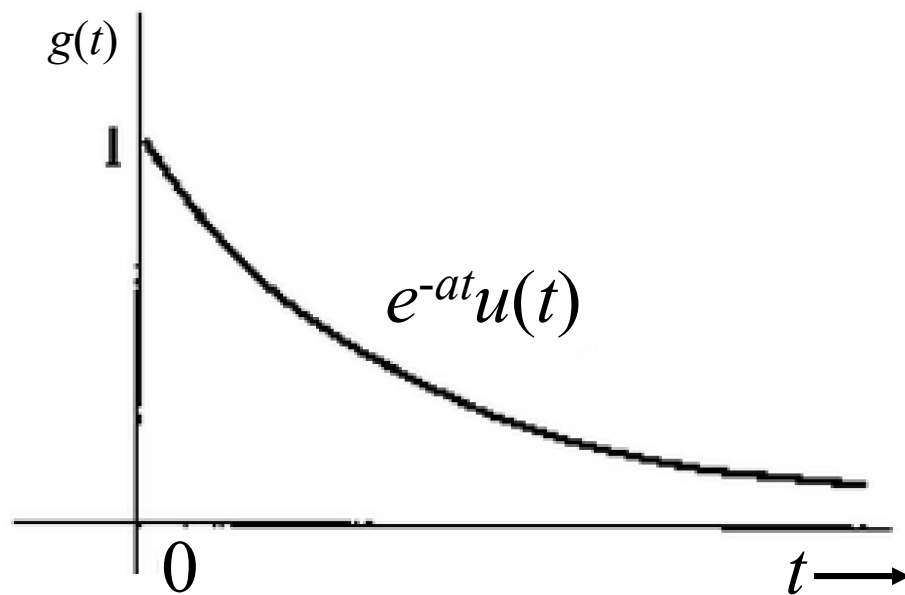
Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Fourier Transform Example

Revisited!

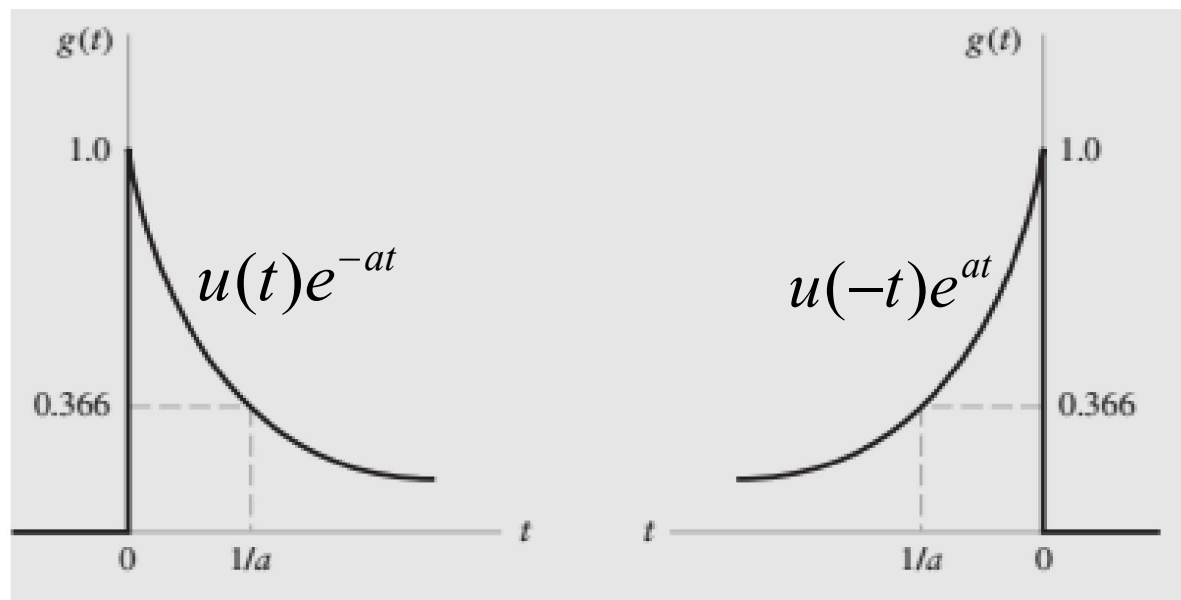


$$\begin{aligned} G(f) &= \frac{-1}{a + j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^\infty \\ &= \frac{1}{a + j2\pi f} \end{aligned}$$

Now as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$ provided $a > 0$

Fourier Transform of Some Useful Functions

Fourier Transform of exponential functions

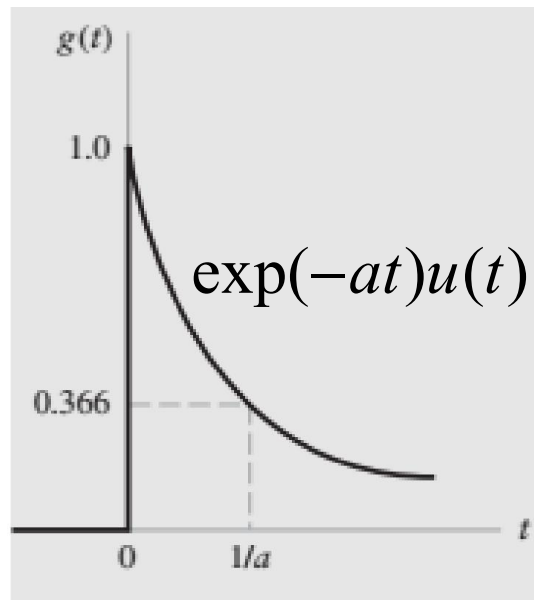


Exponential decaying fnc

Exponential rising fnc

Fourier Transform of Some Useful Functions

Fourier Transform of exponential functions



Exponential decaying fnc

$$\begin{aligned} G(f) &= \int_0^{\infty} e^{-at} e^{-j2\pi ft} dt = \int_0^{\infty} e^{-(a+j2\pi f)t} dt \\ &= \frac{1}{a + j2\pi f} \end{aligned}$$

Fourier Transform Pair

$$\exp(-at)u(t) \Longleftrightarrow \frac{1}{a + j2\pi f}$$

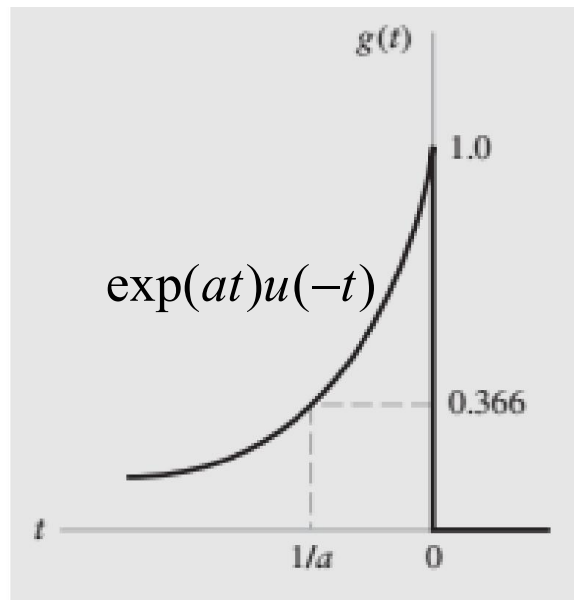
Fourier Transform of Some Useful Functions

Fourier Transform of exponential functions

$$\begin{aligned} G(f) &= \int_{-\infty}^0 e^{at} e^{-j2\pi ft} dt \\ &= \int_0^{\infty} e^{-t(a-j2\pi f)} dt \\ &= \frac{1}{a-j2\pi f} \end{aligned}$$

Fourier Transform Pair

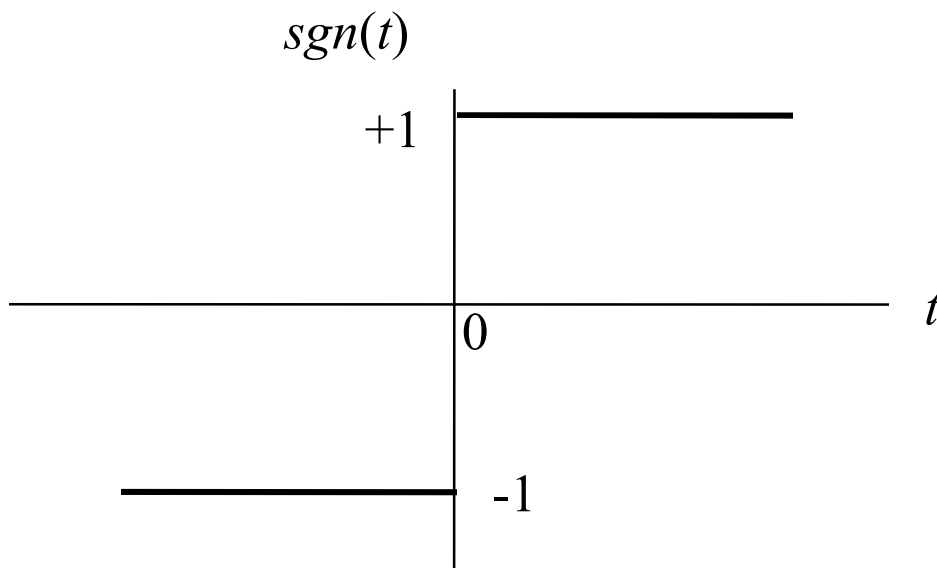
$$\exp(at)u(-t) \rightleftharpoons \frac{1}{a-j2\pi f}$$



Exponential rising fnc

Fourier Transform of Some Useful Functions

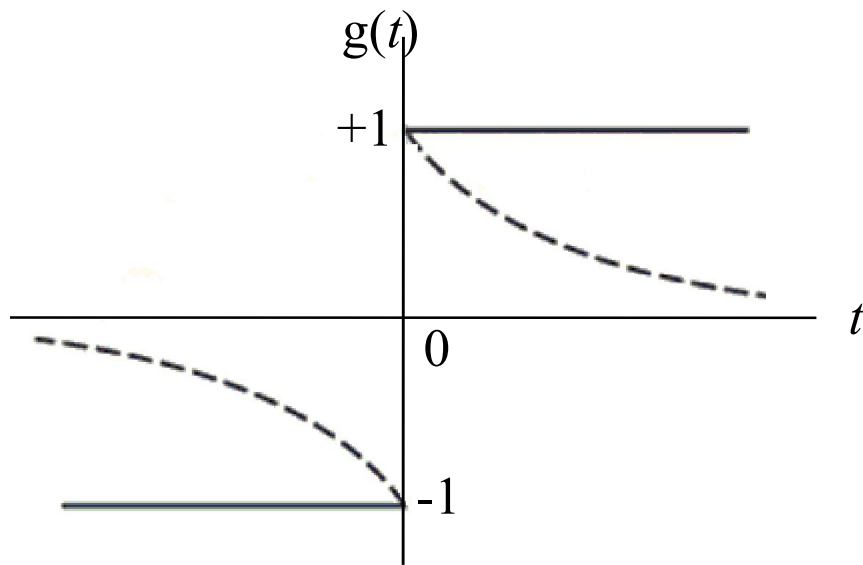
Fourier Transform of $\text{sgn}(t)$ function



$$\text{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

Fourier Transform of Some Useful Functions

Fourier Transform of $\text{sgn}(t)$ function



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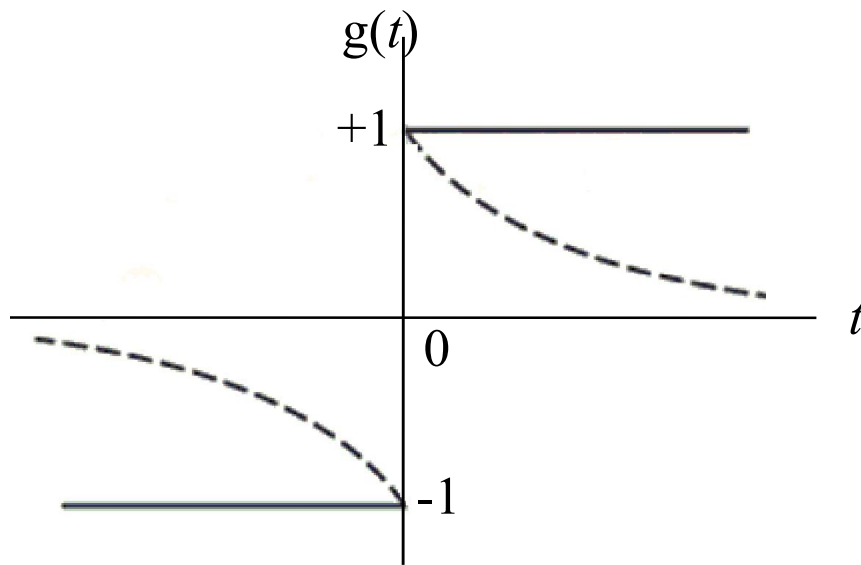
Can be approximated by

$$g(t) = \begin{cases} \exp(-at), & t > 0 \\ 0, & t = 0 \\ -\exp(at), & t < 0 \end{cases}$$

$$g(t) = \exp(-a|t|) \text{sgn}(t)$$

Fourier Transform of Some Useful Functions

Fourier Transform of $\text{sgn}(t)$ function



$$\begin{aligned} \mathcal{F}[\exp(-a|t|) \text{sgn}(t)] &= \frac{1}{a + j2\pi f} - \frac{1}{a - j2\pi f} \\ &= \frac{-j4\pi f}{a^2 + (2\pi f)^2} \end{aligned}$$

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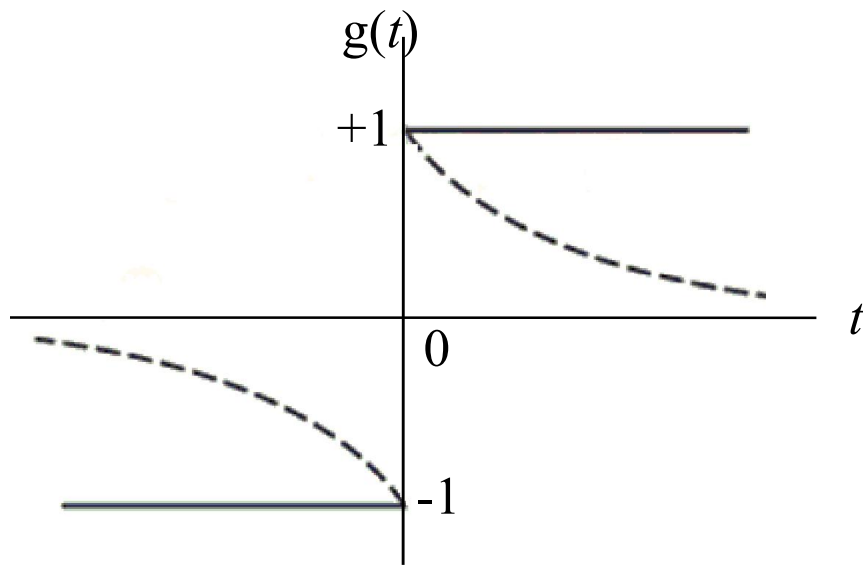
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Fourier Transform of Some Useful Functions

Fourier Transform of $\text{sgn}(t)$ function



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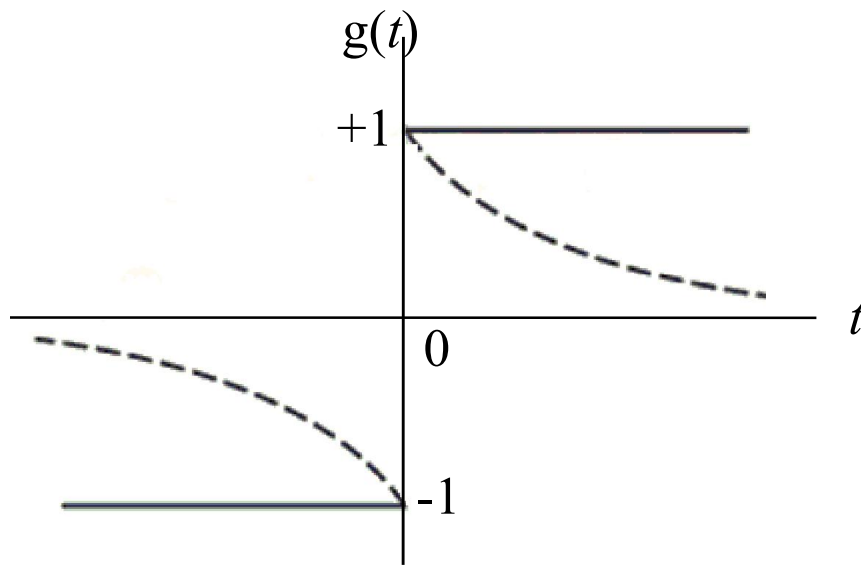
$$\begin{aligned} F[\exp(-a|t|) \text{sgn}(t)] &= \frac{1}{a + j2\pi f} - \frac{1}{a - j2\pi f} \\ &= \frac{-j4\pi f}{a^2 + (2\pi f)^2} \end{aligned}$$

Limiting a to 0

$$\begin{aligned} F[\text{sgn}(t)] &= \lim_{a \rightarrow 0} \frac{-4j\pi f}{a^2 + (2\pi f)^2} \\ &= \frac{1}{j\pi f} \end{aligned}$$

Fourier Transform of Some Useful Functions

Fourier Transform of $\text{sgn}(t)$ function



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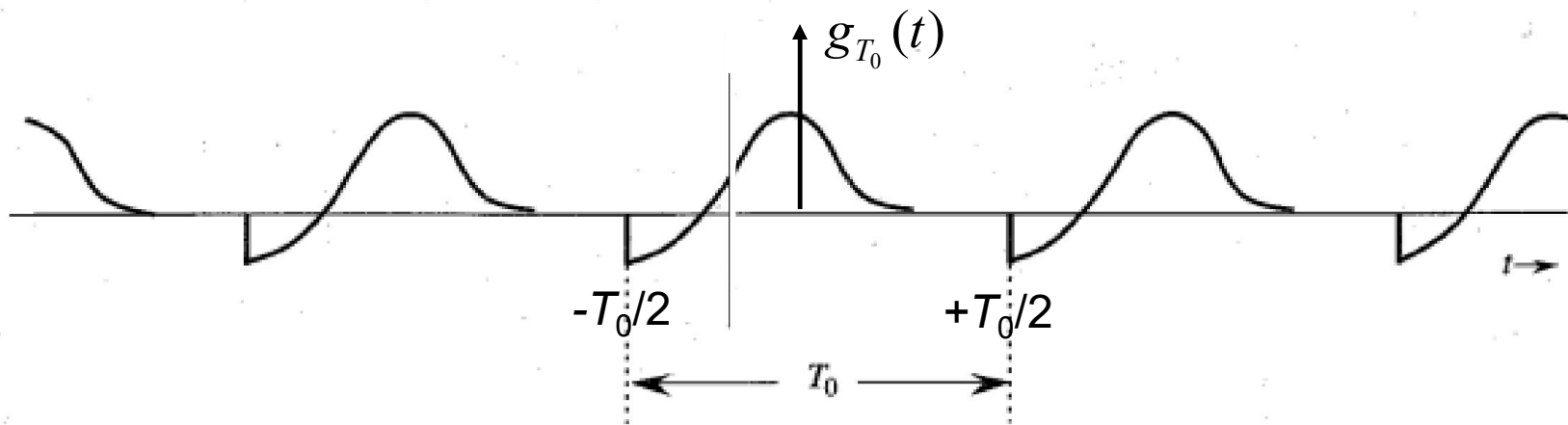
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Fourier Transform Pair

$$\text{sgn}(t) \iff \frac{1}{j\pi f}$$

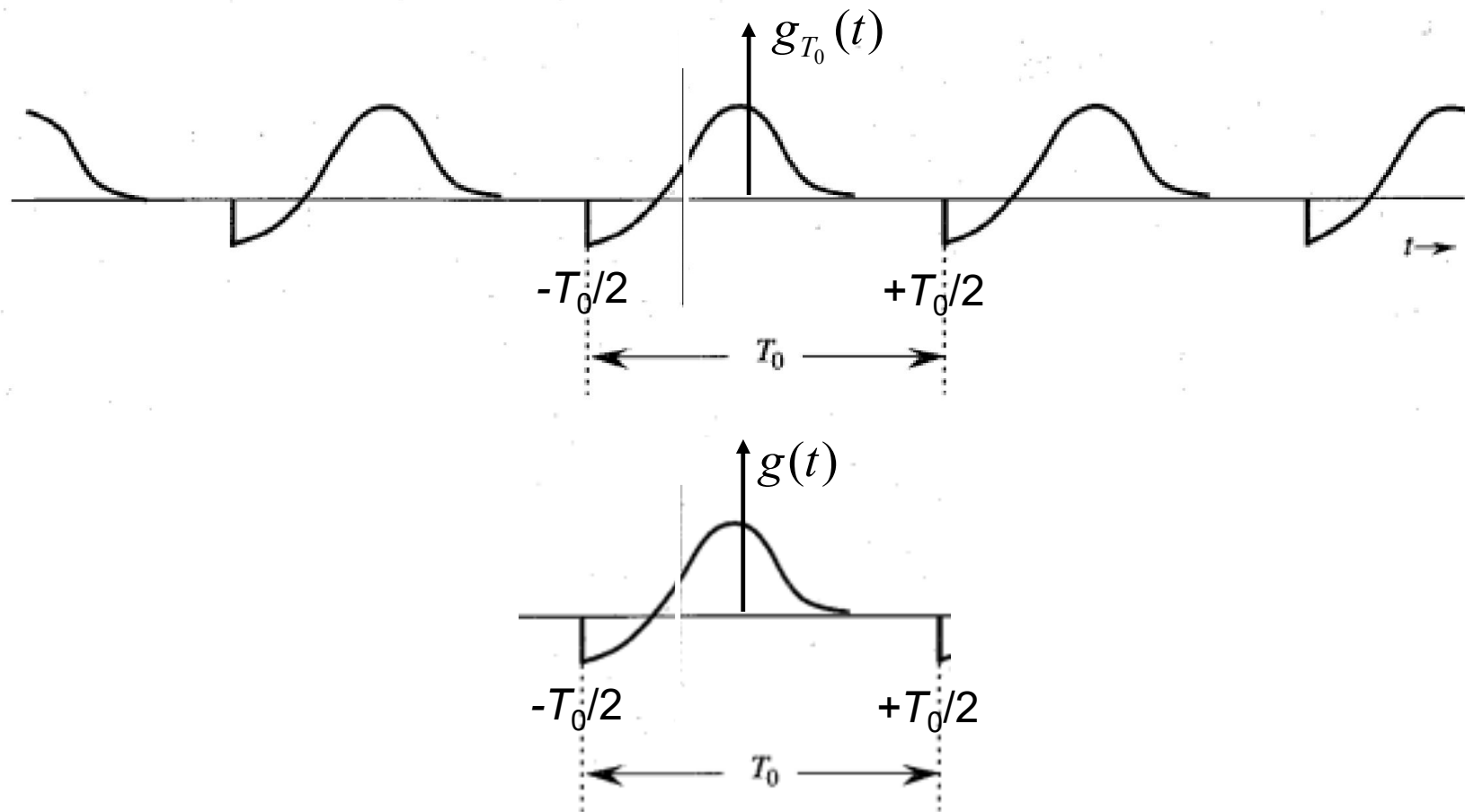
Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



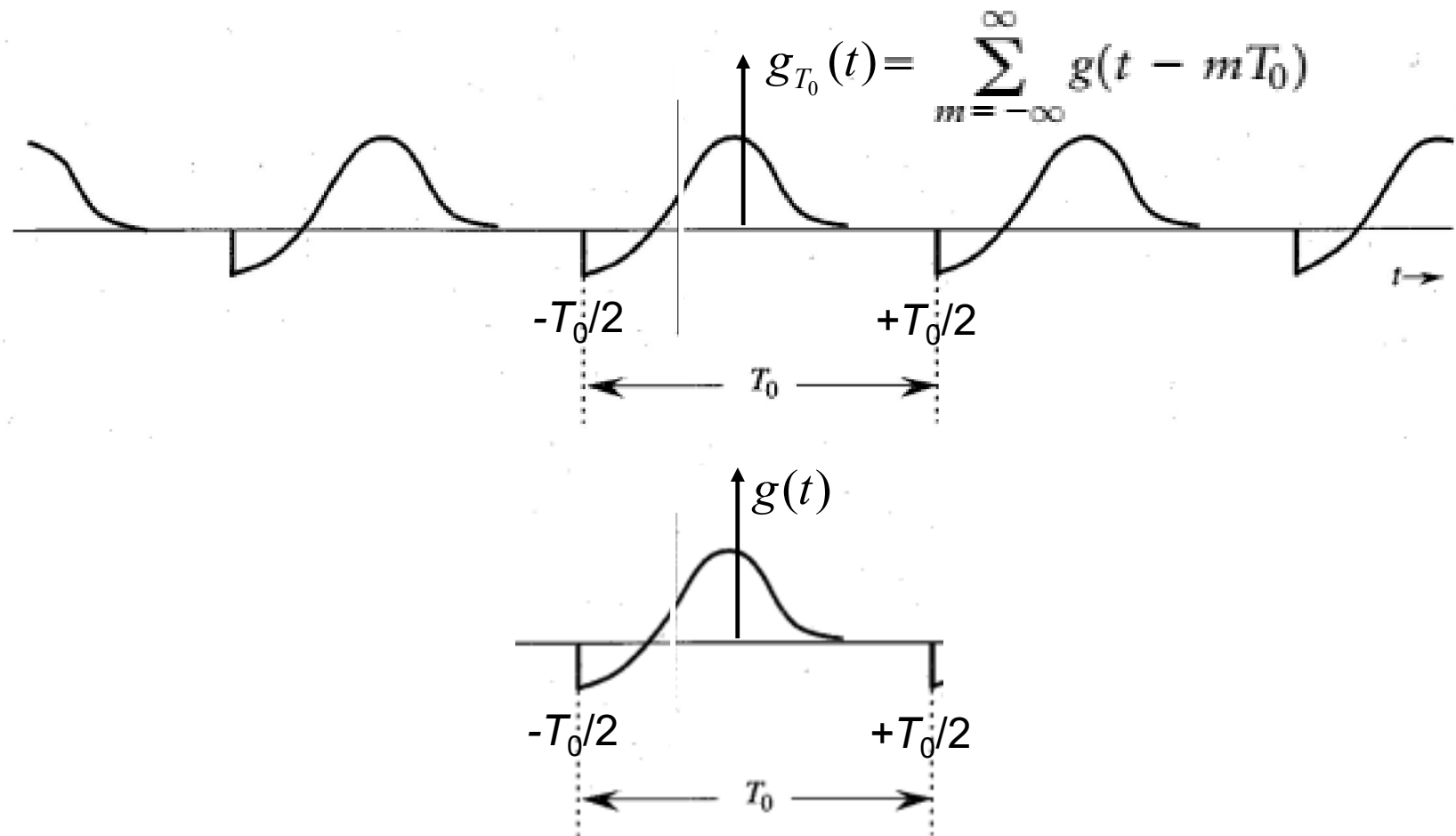
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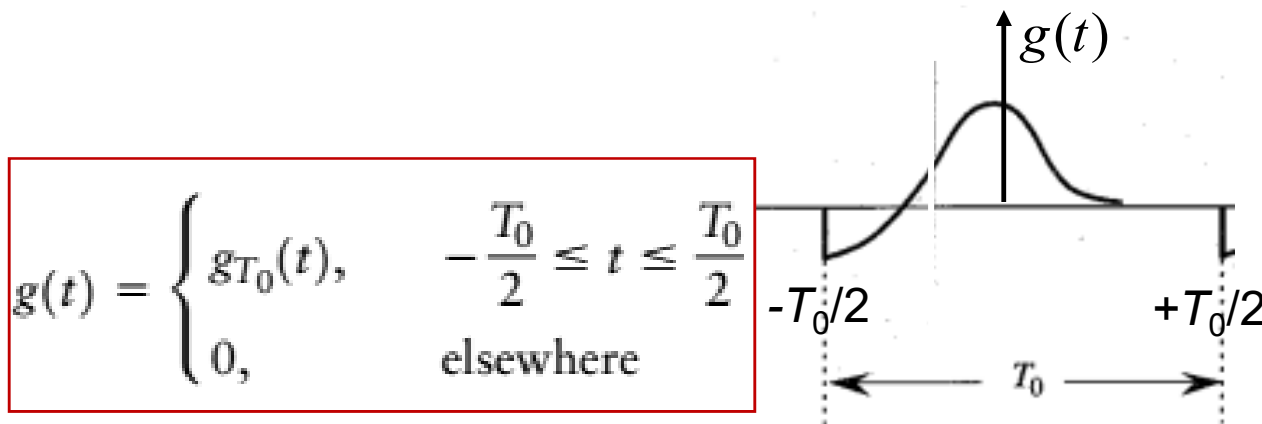
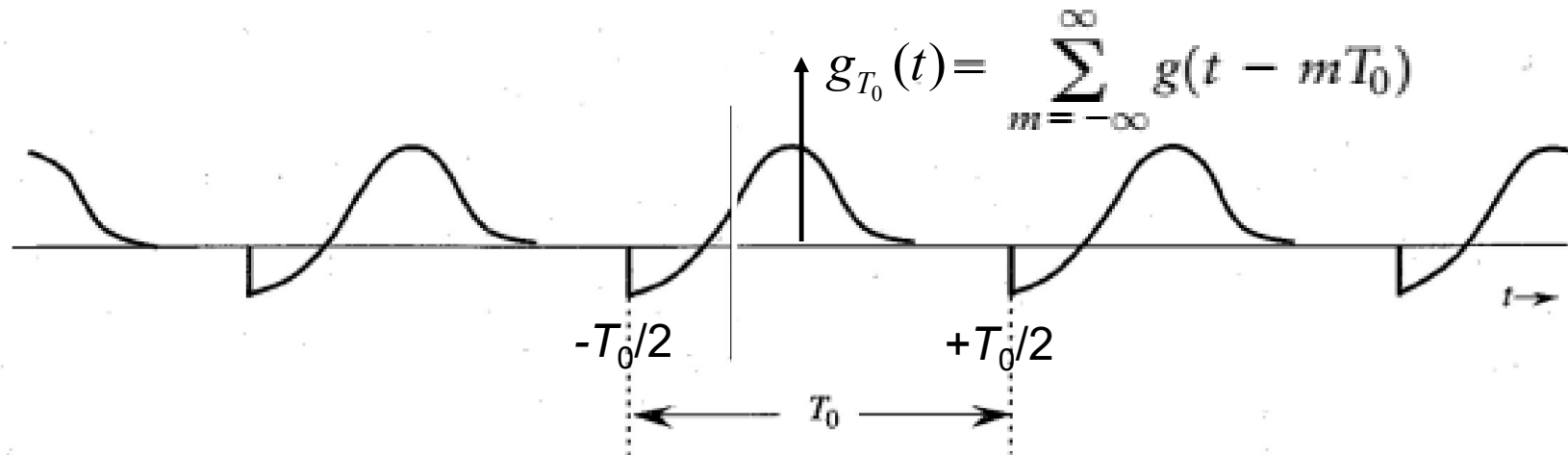
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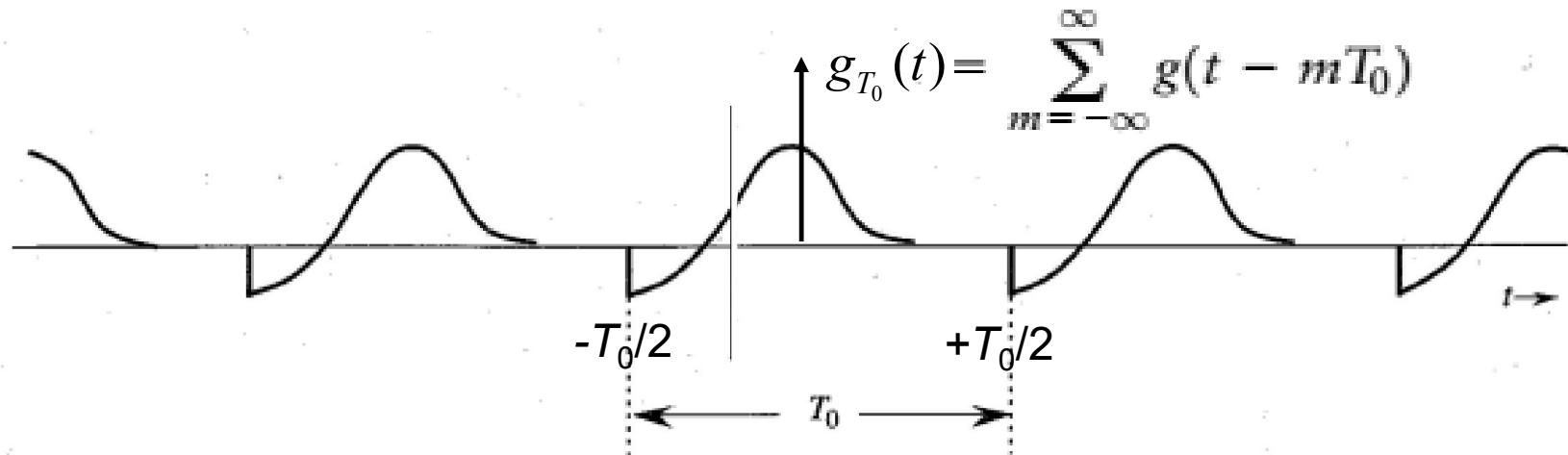
Fourier Transform of Some Useful Functions

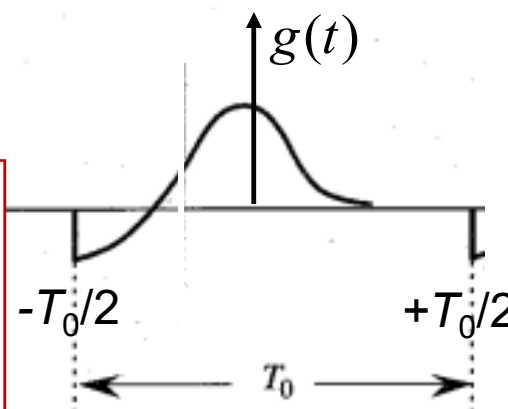
Fourier Transform of a periodic signal $g_{T_0}(t)$



Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



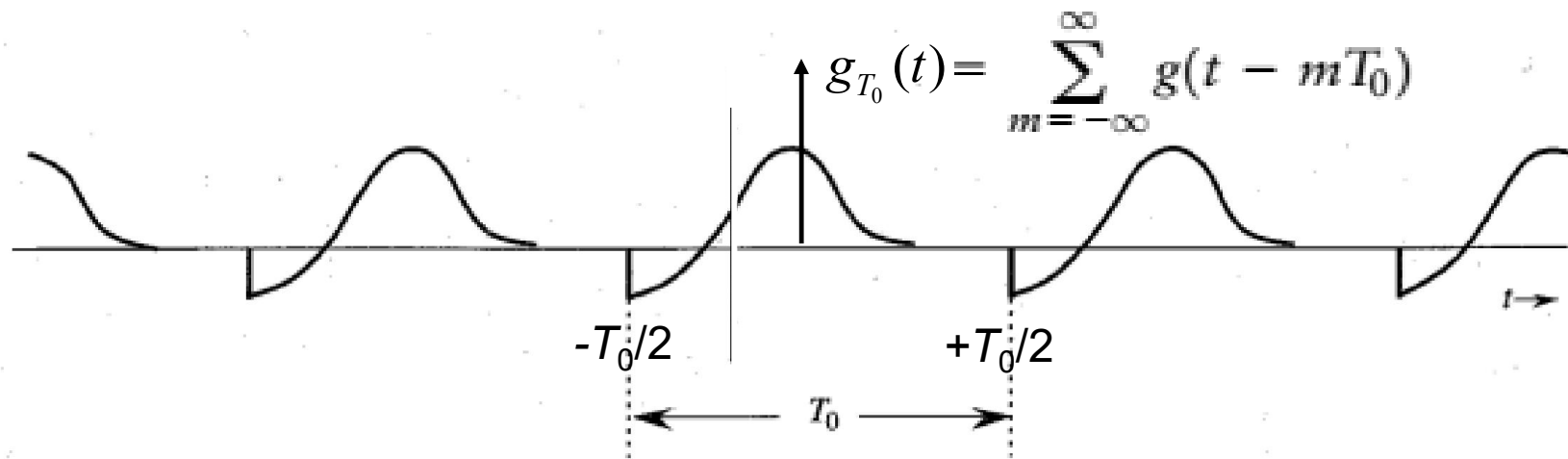
$$g(t) = \begin{cases} g_{T_0}(t), & -\frac{T_0}{2} \leq t \leq \frac{T_0}{2} \\ 0, & \text{elsewhere} \end{cases}$$


The graph shows a single pulse $g(t)$ on a horizontal axis. The pulse is centered at $t = 0$ and has a width of T_0 . The period T_0 is indicated by the distance between two consecutive pulses.

Assume that
 $\mathfrak{F}[g(t)] = G(f)$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



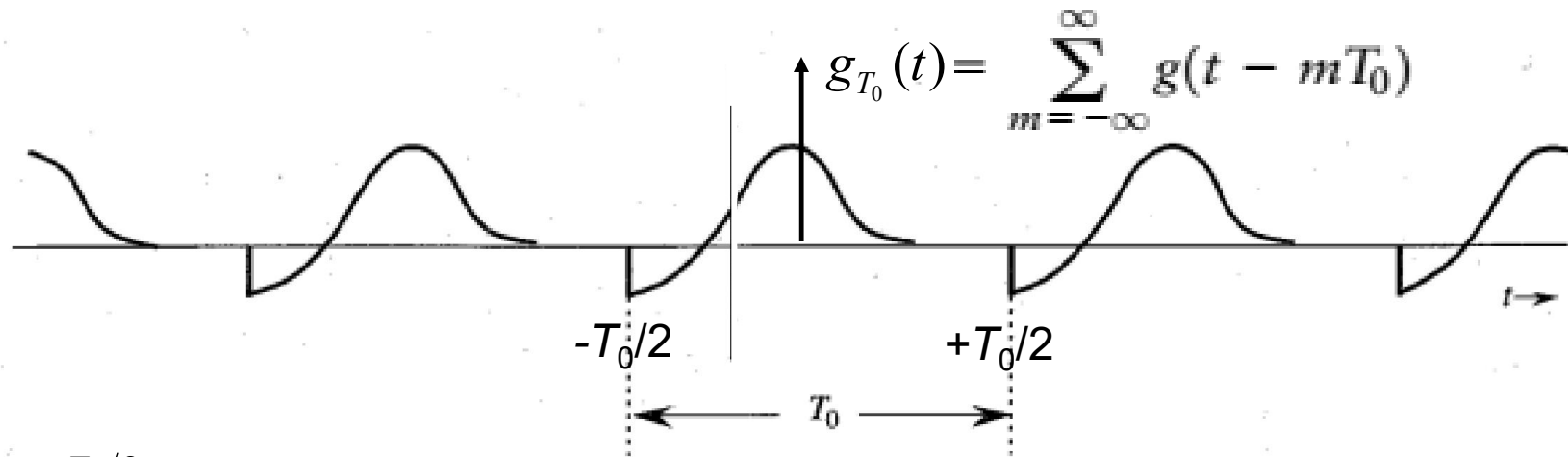
As it is periodic,

$$g_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} c_n \exp(j2\pi n f_0 t)$$

where, $c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt$ and $f_0 = \frac{1}{T_0}$

Fourier Transform of Some Useful Functions

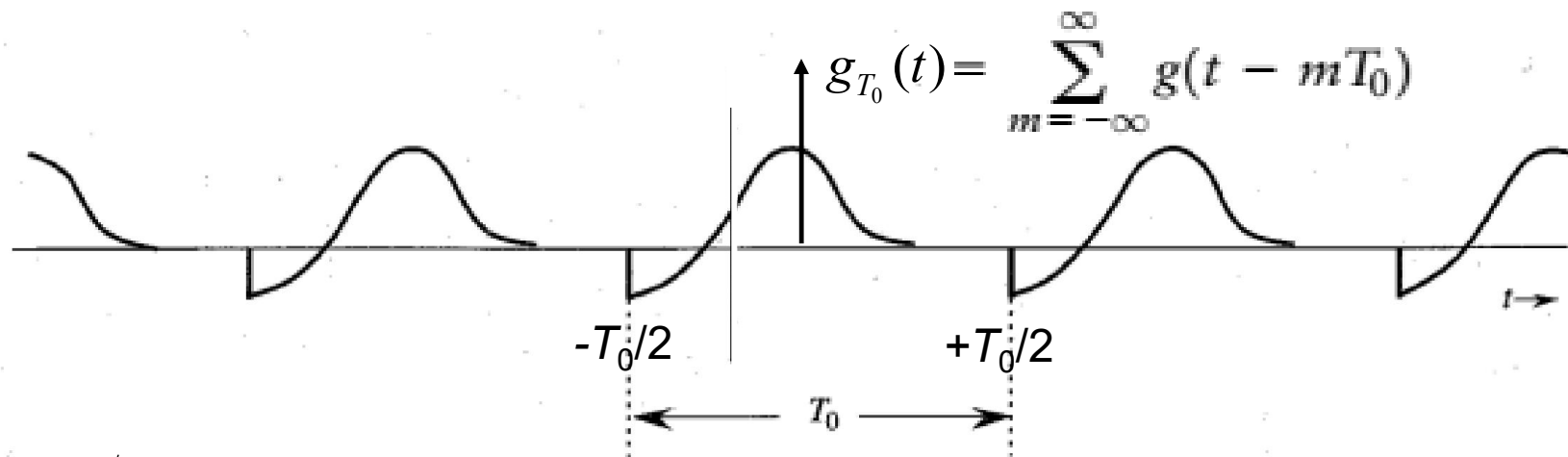
Fourier Transform of a periodic signal $g_{T_0}(t)$

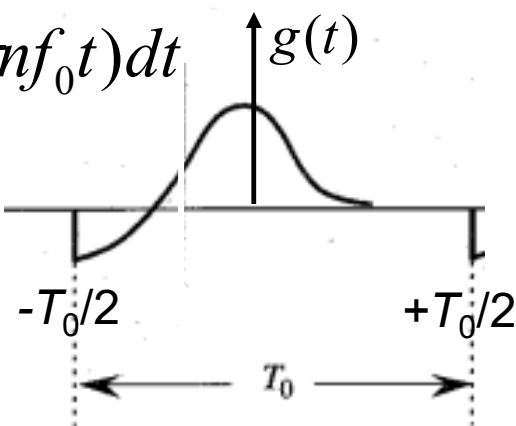


$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$

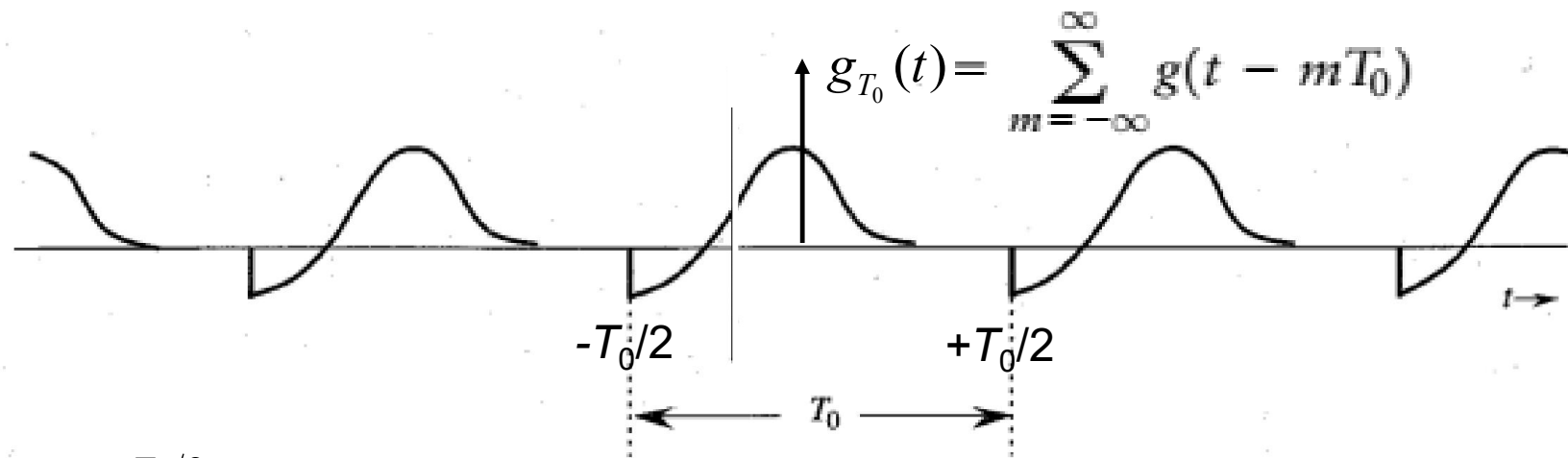


$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt$$


The graph shows a single period $g(t)$ of the periodic signal $g_{T_0}(t)$ plotted against time t . The period T_0 is indicated by a double-headed arrow between two consecutive peaks. The time axis is labeled t . The period T_0 is also indicated by a double-headed arrow between two consecutive peaks. The period T_0 is also indicated by a double-headed arrow between two consecutive peaks.

Fourier Transform of Some Useful Functions

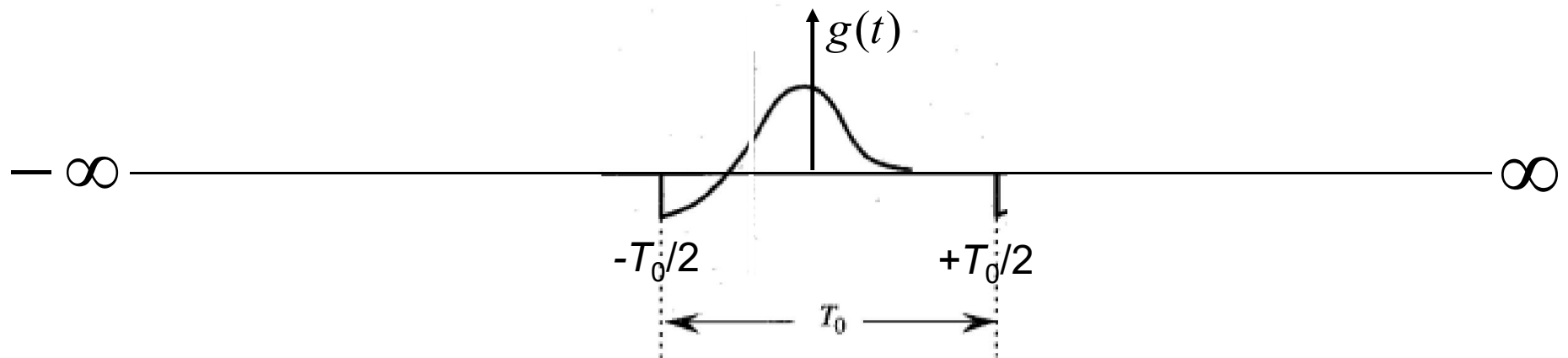
Fourier Transform of a periodic signal $g_{T_0}(t)$



$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g(t) \exp(-j2\pi n f_0 t) dt$$

Fourier Transform of Some Useful Functions

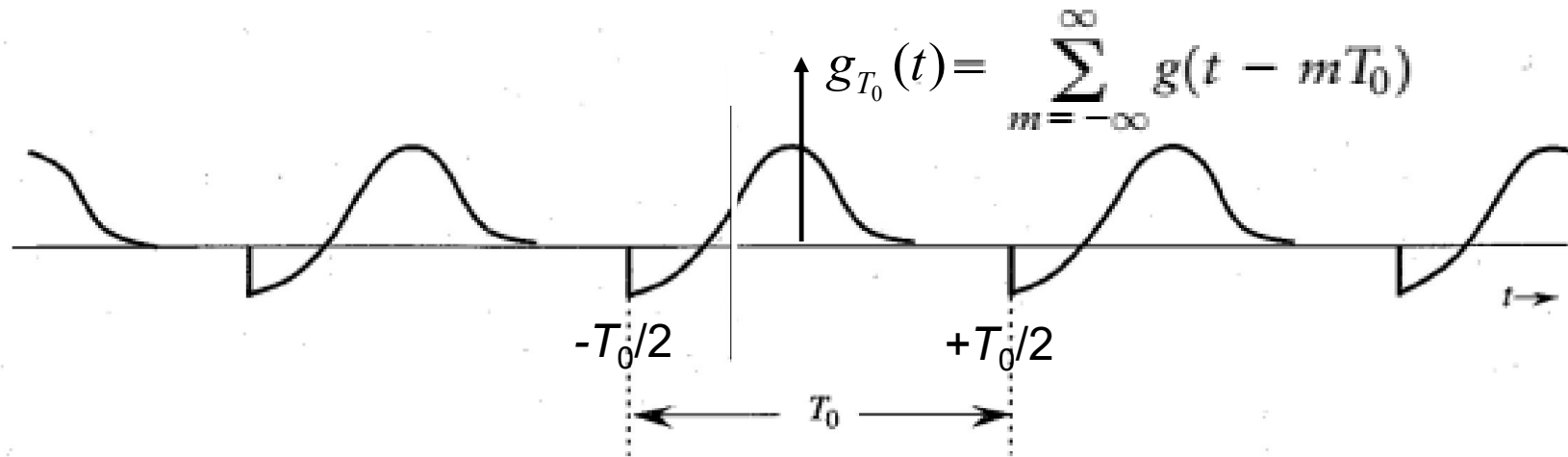
Fourier Transform of a periodic signal $g_{T_0}(t)$



$$\begin{aligned} c_n &= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g(t) \exp(-j2\pi n f_0 t) dt \\ &= f_0 \int_{-\infty}^{\infty} g(t) \exp(-j2\pi n f_0 t) dt \\ &= f_0 G(n f_0) \end{aligned}$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



$$c_n = \frac{1}{T_0} \sum_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt$$

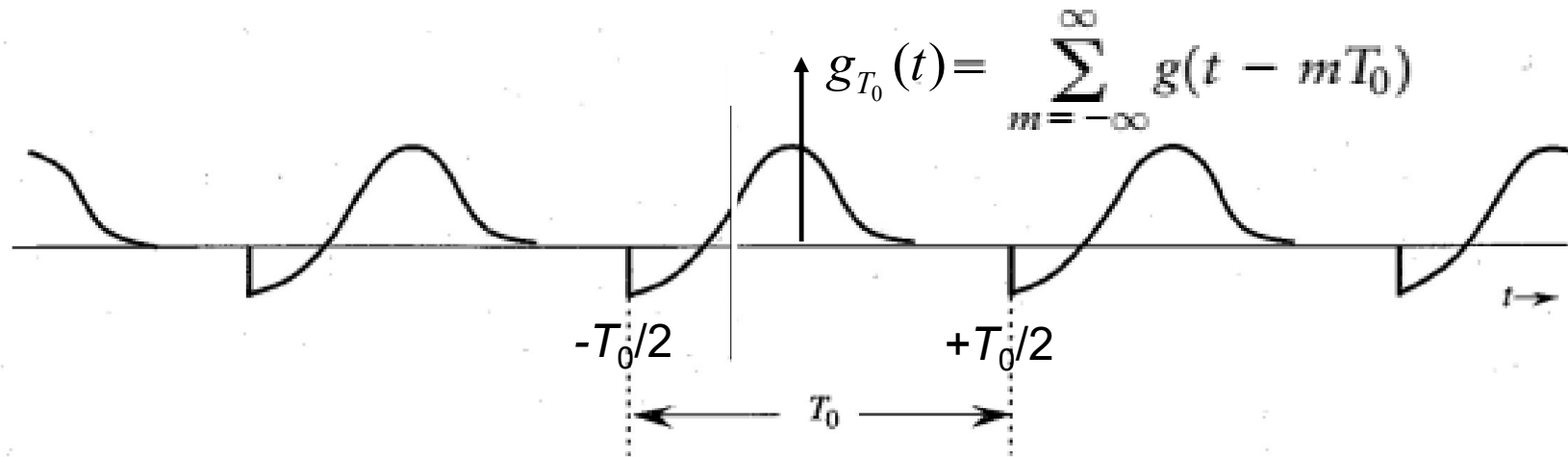
$$= f_0 \int_{-\infty}^{\infty} g(t) \exp(-j2\pi n f_0 t) dt$$

$$= f_0 G(nf_0)$$

where $G(nf_0)$ is FT of $g(t)$ evaluated at $f = nf_0$

Fourier Transform of Some Useful Functions

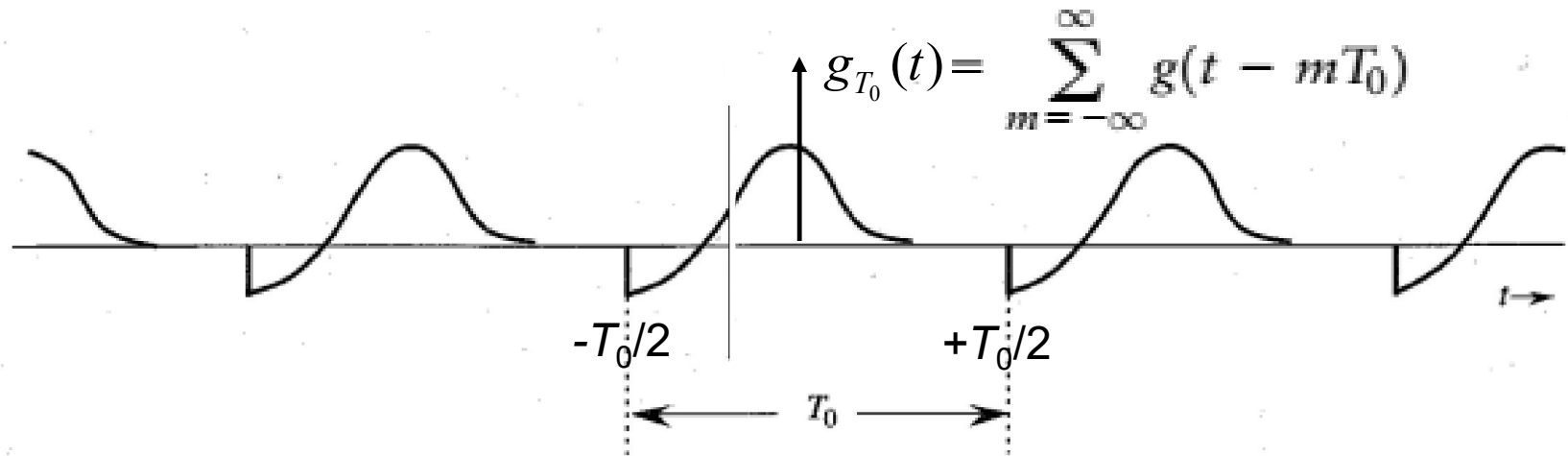
Fourier Transform of a periodic signal $g_{T_0}(t)$



$$g_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} c_n \exp(j2\pi n f_0 t) \quad \text{and we found, } c_n = f_0 G(nf_0)$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$

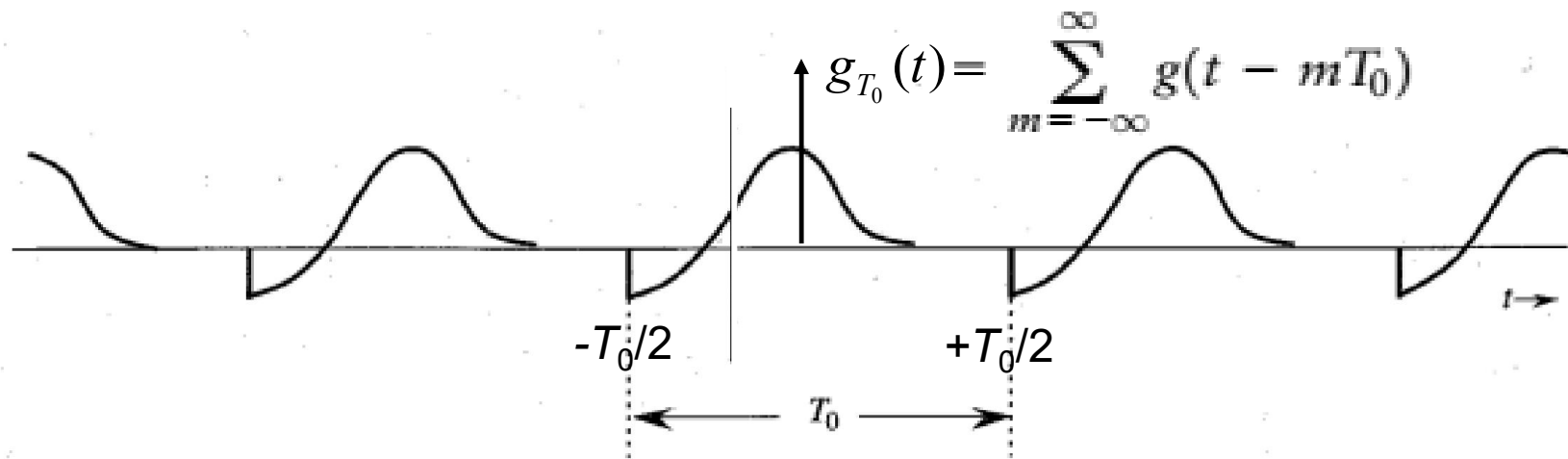


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$$= f_0 \sum_{n=-\infty}^{\infty} G(nf_0) \exp(j2\pi n f_0 t)$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



$$\begin{aligned}\mathfrak{T}[g_{T_0}(t)] &= \mathfrak{T}\left[f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \exp(j2\pi nf_0 t)\right] = f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \mathfrak{T}[\exp(j2\pi nf_0 t)] \\ &= f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \delta(f - nf_0)\end{aligned}$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$

$$\mathfrak{F}[g_{T_0}(t)] = f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \delta(f - nf_0) \quad \mathfrak{F}[g(t)] = G(f)$$

- $g(t)$ is a single period of $g_{T_0}(t)$ but its spectrum is continuous
- Spectrum of $g_{T_0}(t)$ is discrete
- Spectrum of $g_{T_0}(t)$ consists of
 - Impulse functions occurring at integer multiples of f_0
 - each impulse function is weighted by a factor equal to the corresponding value of $G(nf_0)$

Fourier Transform of Some Useful Functions

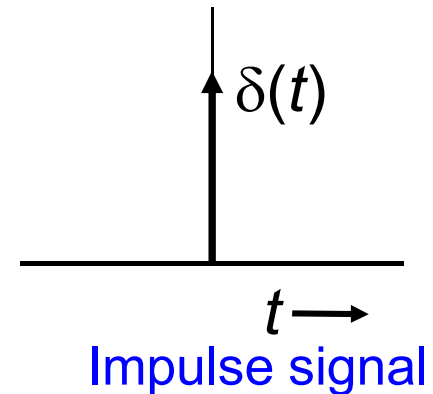
Fourier Transform of an impulse train $\delta_{T_0}(t)$

- $\delta_{T_0}(t)$ is a periodic signal with period T_0

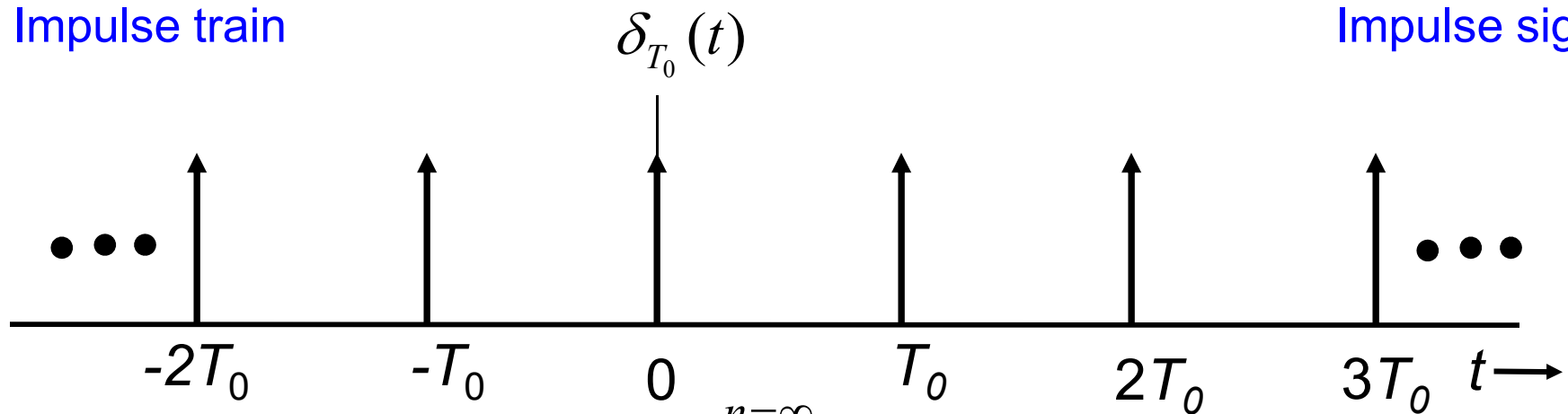
Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

- $\delta_{T_0}(t)$ is a periodic signal with period T_0



Impulse train

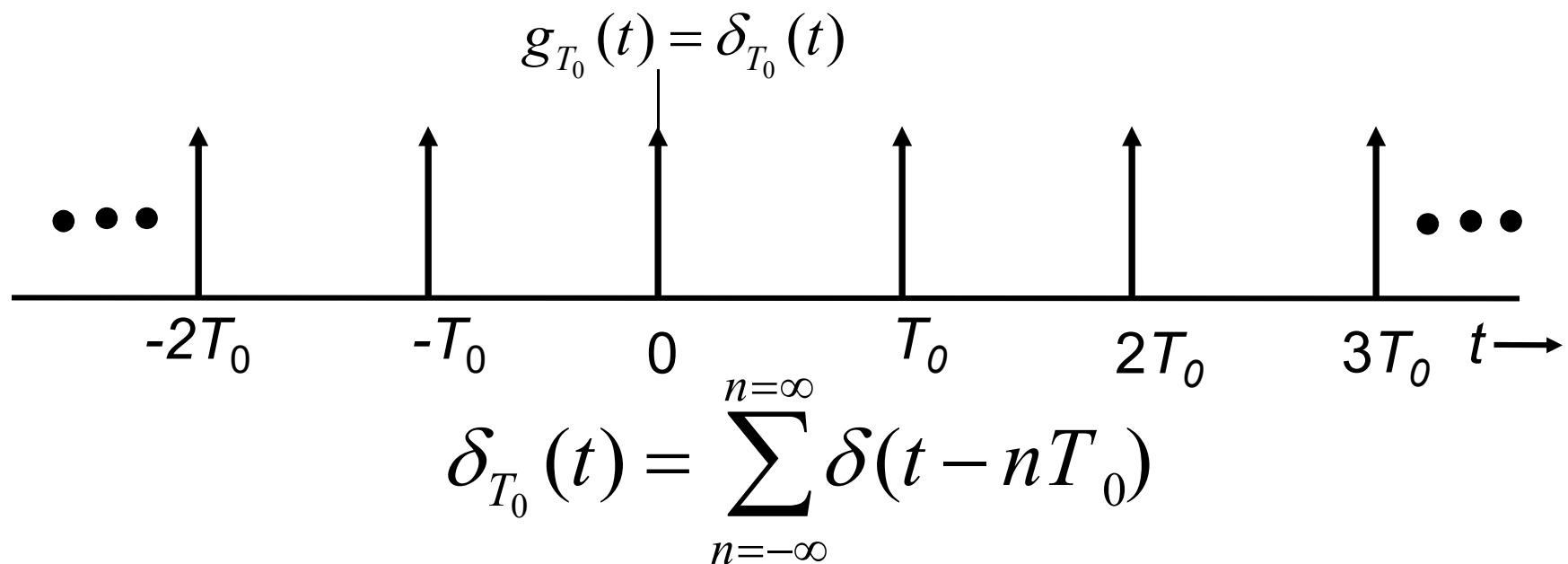


Equation of an
Impulse train

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} \delta(t - nT_0)$$

Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

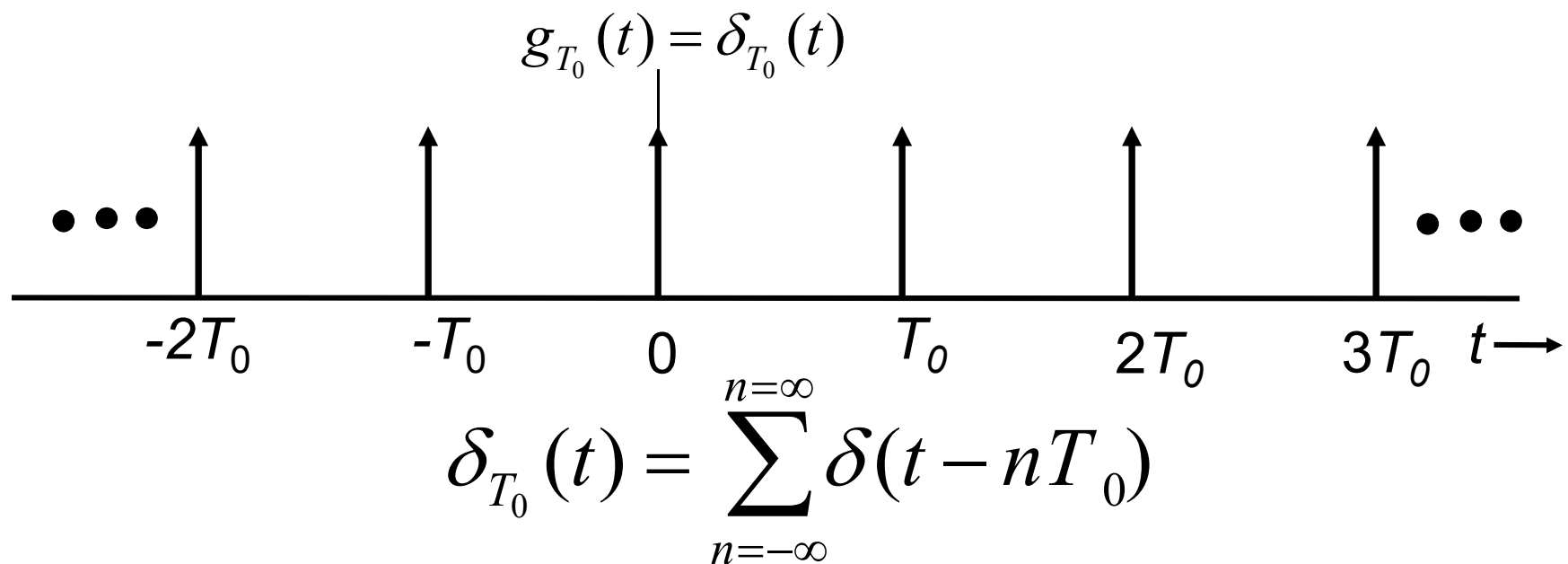


Comparing with the previous result

$$\mathfrak{F}[g_{T_0}(t)] = f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \delta(f - nf_0) \qquad \mathfrak{F}[g(t)] = G(f)$$

Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

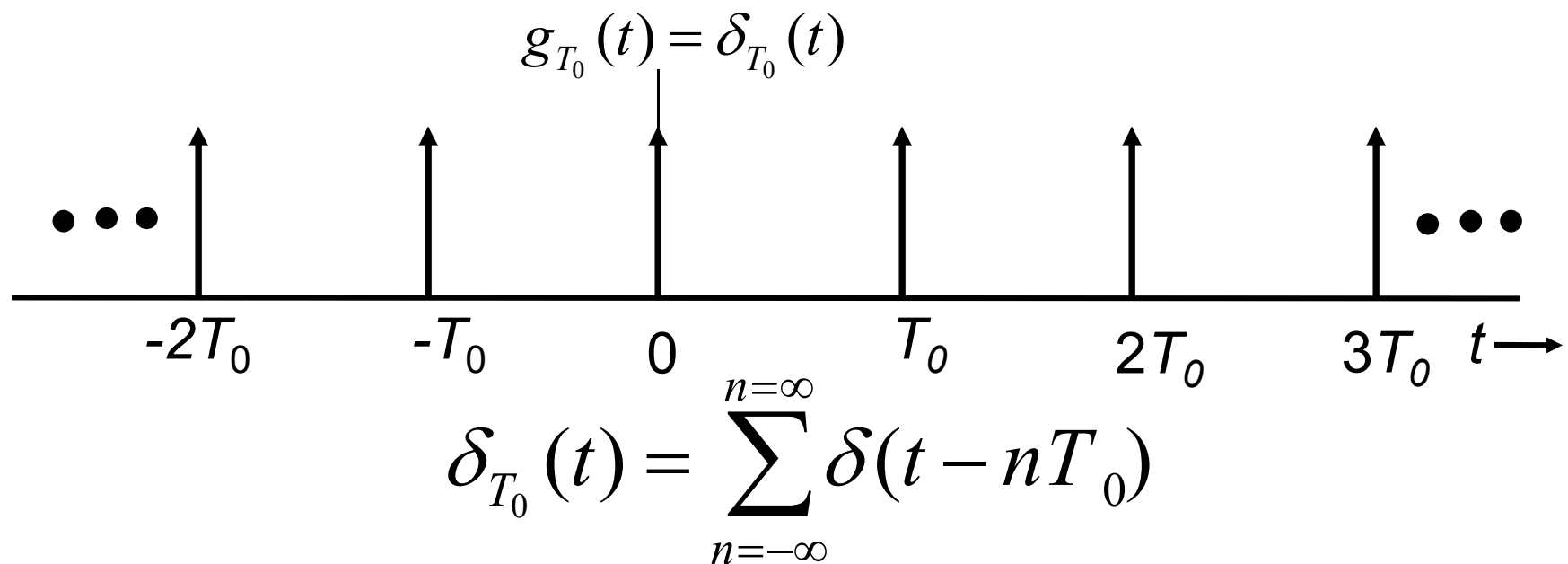


Comparing with the previous result

$$G(f) = \mathfrak{F}[g(t)] = \mathfrak{F}[\delta(t)] = 1$$

Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

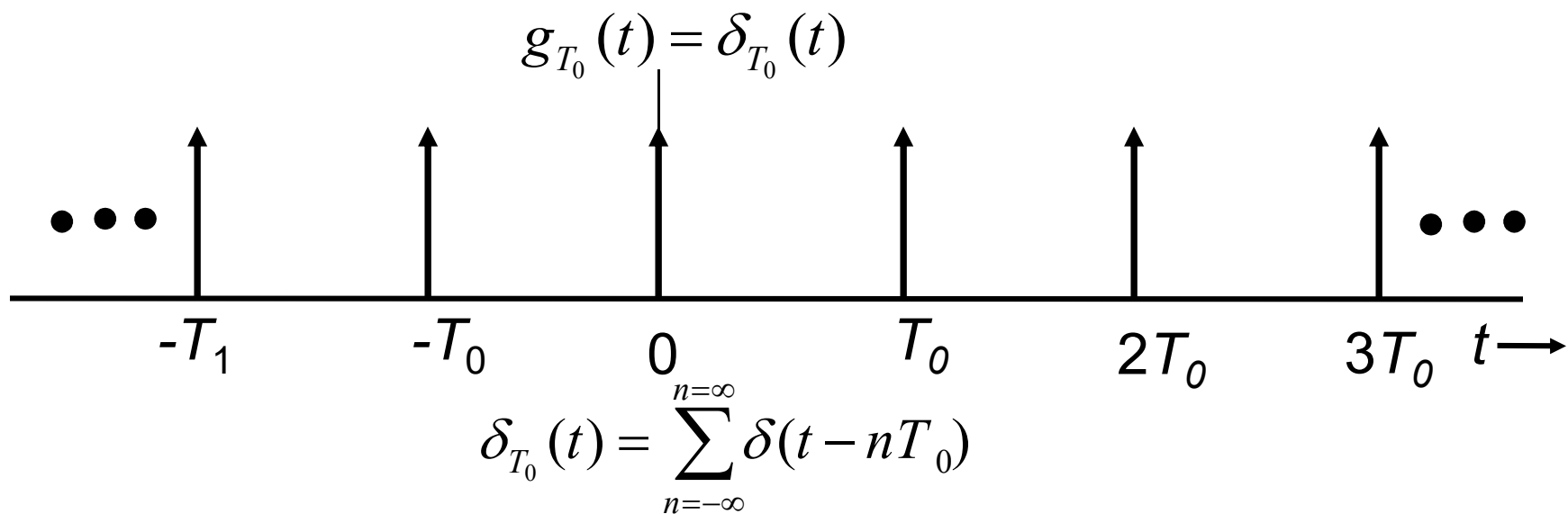


Comparing with the previous result

$$G(f) = \mathfrak{F}[g(t)] = \mathfrak{F}[\delta(t)] = 1 = G(nf_0)$$

Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

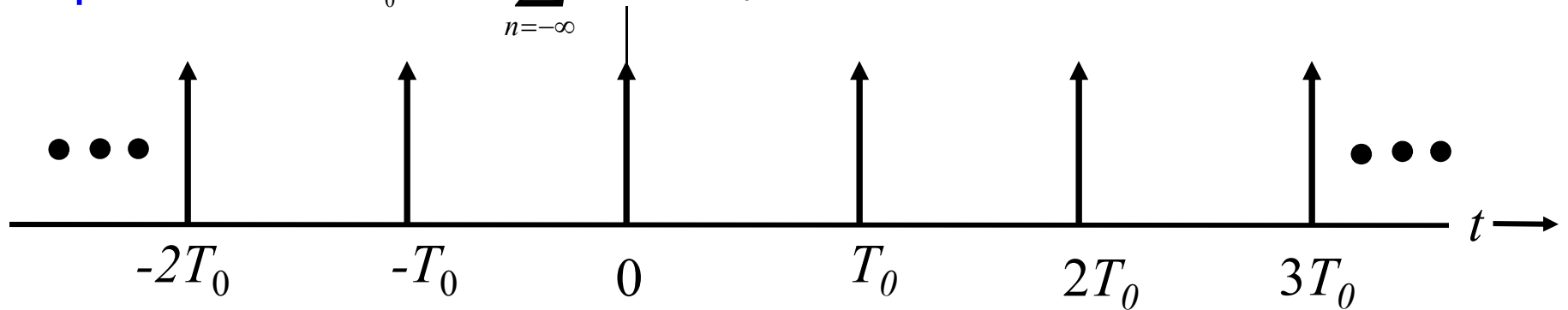


Comparing with the previous result

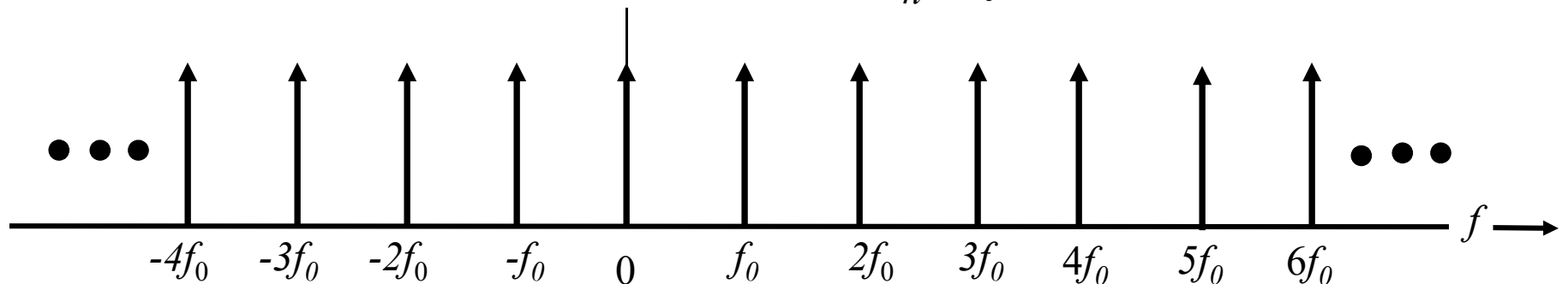
$$\mathfrak{F}[\delta_{T_0}(t)] = f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \delta(f - nf_0) = f_0 \sum_{n=-\infty}^{n=\infty} \delta(f - nf_0)$$

Fourier Transform of Some Useful Functions

Impulse train $\delta_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} \delta(t - nT_0)$



FT of Impulse train $\mathfrak{F}[\delta_{T_0}(t)] = f_0 \sum_{n=-\infty}^{n=\infty} \delta(f - nf_0)$



Fourier Transform of Some Useful Functions

	$g(t)$	$G(f)$	
1	$e^{-at}u(t)$	$\frac{1}{a + j2\pi f}$	$a > 0$
2	$e^{at}u(-t)$	$\frac{1}{a - j2\pi f}$	$a > 0$
3	$e^{-a t }$	$\frac{2a}{a^2 + (2\pi f)^2}$	$a > 0$
4	$te^{-at}u(t)$	$\frac{1}{(a + j2\pi f)^2}$	$a > 0$
5	$t^n e^{-at}u(t)$	$\frac{n!}{(a + j2\pi f)^{n+1}}$	$a > 0$
6	$\delta(t)$	1	
7	1	$\delta(f)$	
8	$e^{j2\pi f_0 t}$	$\delta(f - f_0)$	
9	$\cos 2\pi f_0 t$	$0.5 [\delta(f + f_0) + \delta(f - f_0)]$	
10	$\sin 2\pi f_0 t$	$j0.5 [\delta(f + f_0) - \delta(f - f_0)]$	

	$g(t)$	$G(f)$
17	$\Pi\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc}(\pi f \tau)$
18	$2B \operatorname{sinc}(2\pi Bt)$	$\Pi\left(\frac{f}{2B}\right)$
19	$\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2} \operatorname{sinc}^2\left(\frac{\pi f \tau}{2}\right)$

Properties of Fourier Transform

Time-Frequency duality

Analysis equation:

$$G(f) = \int_{-\infty}^{\infty} g(t) \exp(-j2\pi ft) dt$$

Time-domain
description:
 $g(t)$

Frequency-domain
description:
 $G(f)$

Synthesis equation:

$$g(t) = \int_{-\infty}^{\infty} G(f) \exp(j2\pi ft) df$$

Properties of Fourier Transform

Time-Frequency duality

Analysis equation:

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- Difference is minor
- Remarkably similar

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Properties of Fourier Transform

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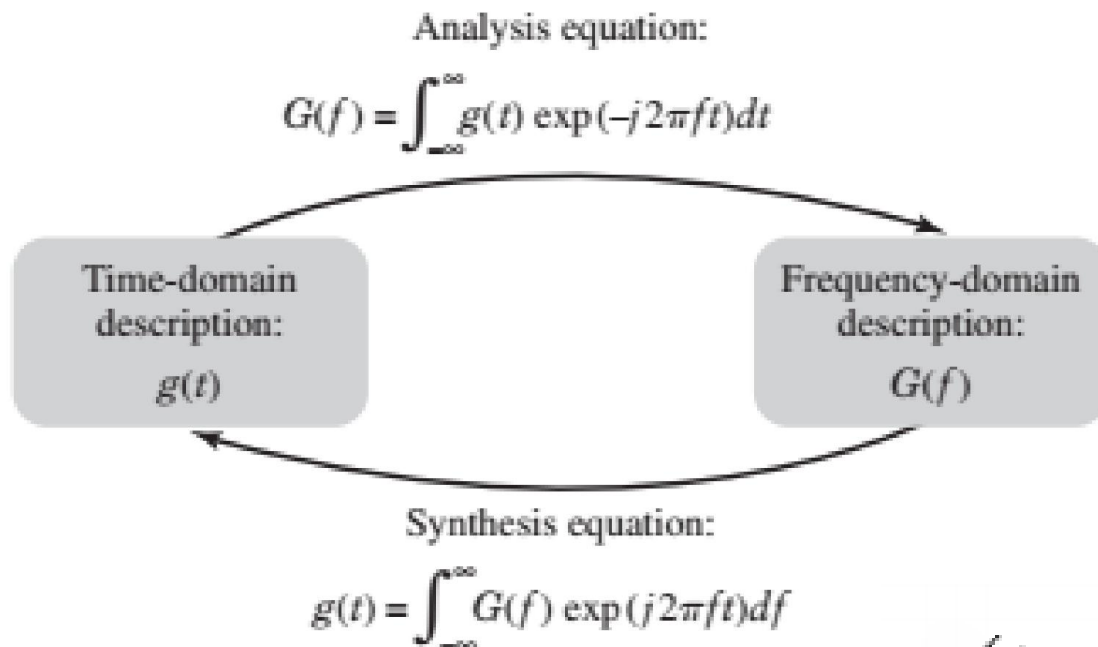
Synthesis equation:

$$g(t) = \int_{-\infty}^{\infty} G(f) \exp(j2\pi ft) df$$

- For any relationship, dual exists by interchanging roles of $g(t)$ and $G(f)$

Properties of Fourier Transform

Time-Frequency duality



Let, $g(t) \Leftrightarrow G(f)$

$$g(t - t_0) \Leftrightarrow G(f) e^{-j2\pi f t_0}$$

means

$$g(t) e^{j2\pi f_0 t} \Leftrightarrow G(f - f_0)$$

Properties of Fourier Transform

Duality property

$$\text{If } g(t) \Leftrightarrow G(f)$$

$$\text{then } G(t) \Leftrightarrow g(-f)$$

Properties of Fourier Transform

Duality property

Proof

If $g(t) \Leftrightarrow G(f)$

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

then $G(t) \Leftrightarrow g(-f)$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} G(x) e^{j2\pi xt} dx$$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} G(x) e^{j2\pi xt} dx$$

$$g(-t) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xt} dx \quad \text{Replaced } t \text{ by } -t$$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

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Proof

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$$g(-f) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xf} dx \quad \text{Replaced } t \text{ by } f$$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

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Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

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This means, $g(-f) = \mathfrak{I}[G(t)]$

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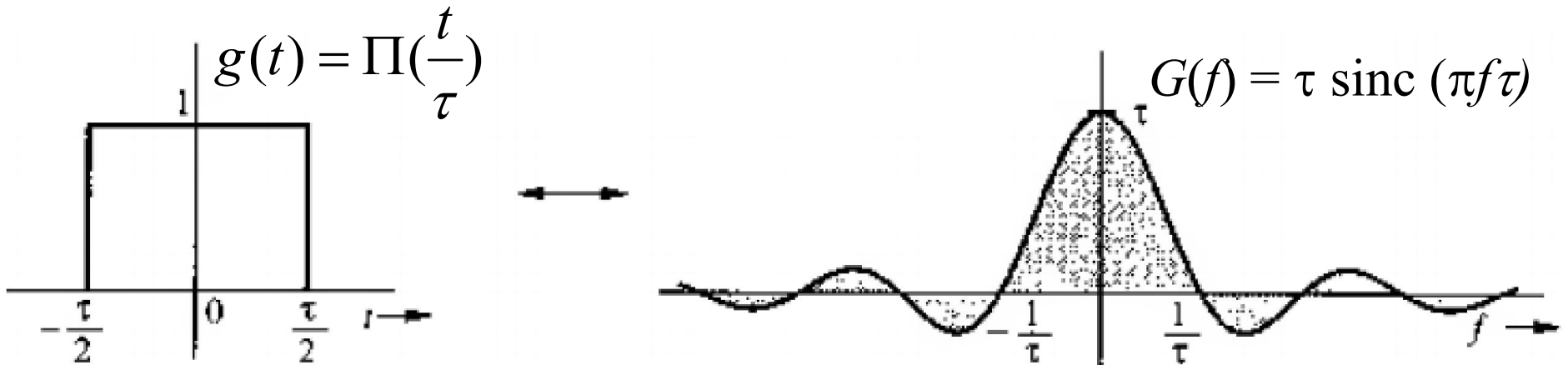
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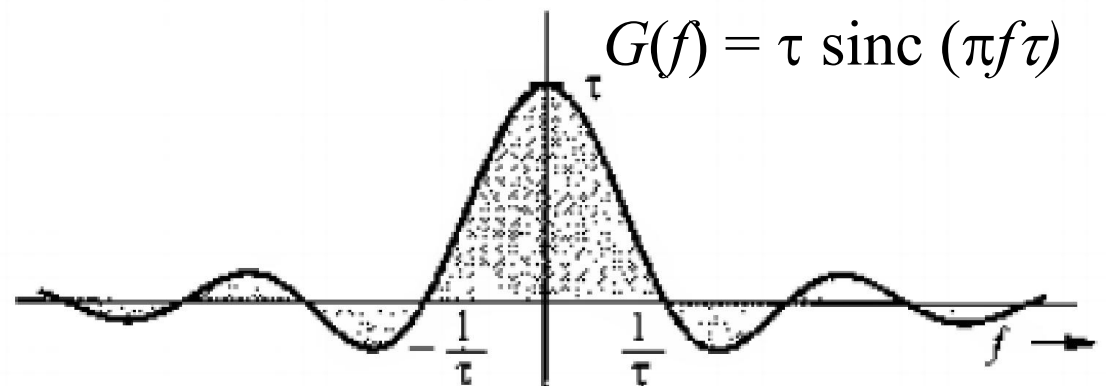
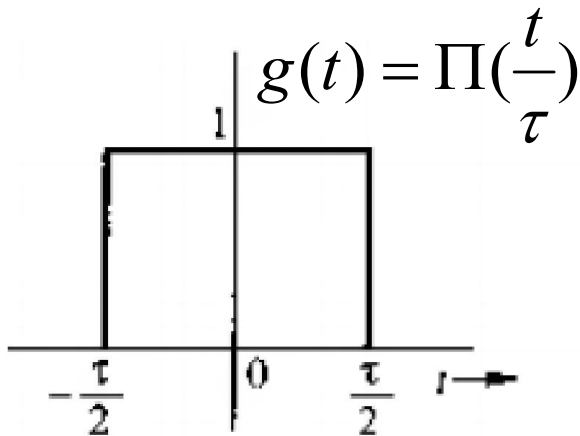
Properties of Fourier Transform

Use of Duality property $G(t) \Leftrightarrow g(-f)$



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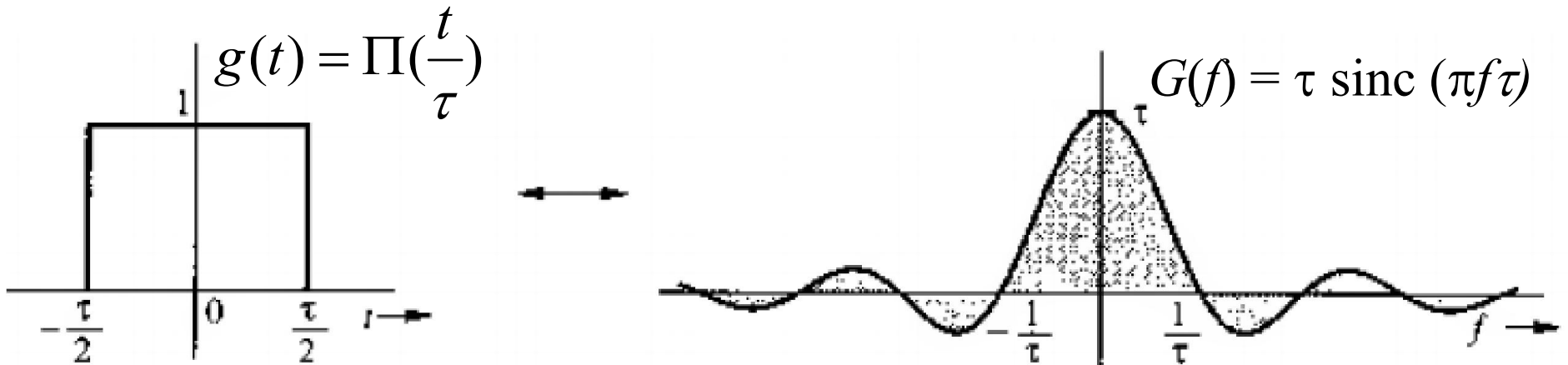


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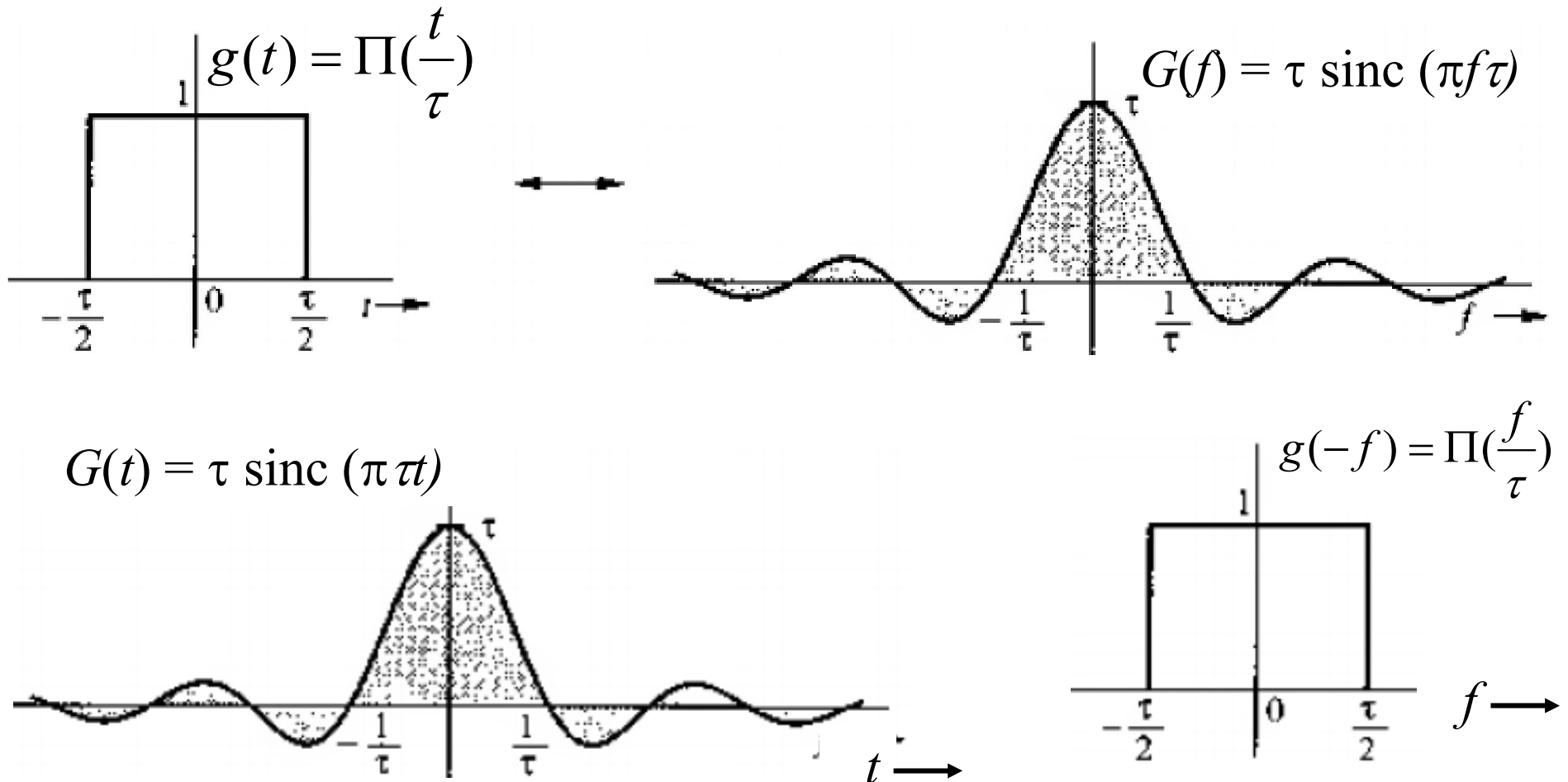


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Properties of Fourier Transform

Use of Duality property $G(t) \Leftrightarrow g(-f)$



Properties of Fourier Transform

Time scaling or dilation
property

$$g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$$

Properties of Fourier Transform

Time scaling or dilation property $g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$

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Properties of Fourier Transform

Time scaling or dilation property $g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$

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$$\text{Therefore, } g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$$

Properties of Fourier Transform

Significance of Time
scaling property

$$g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$$

- If $a > 1$, $g(at)$ means **compression in time** which **increases the signal frequency**
- Compression in time results in spectral expansion by same factor

Properties of Fourier Transform

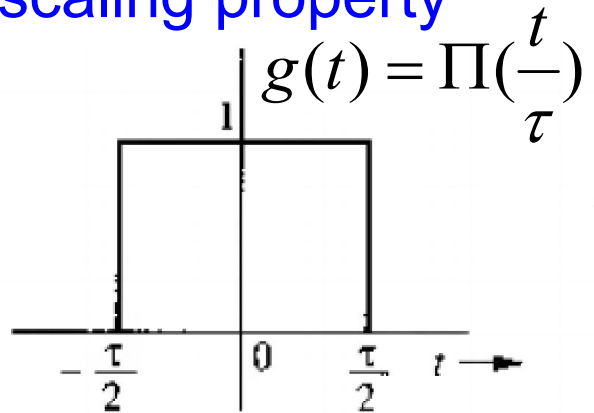
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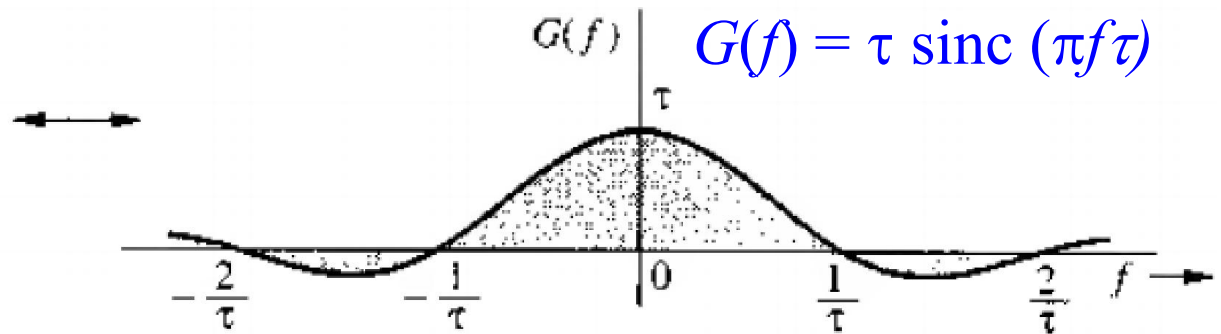
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- Compression in time results in spectral expansion by same factor
- Expansion in time ($a < 1$) results in spectral compression
- If $g(t)$ is wider, spectrum is narrower
- Bandwidth (spectrum) is inversely proportional to signal duration (width)

Properties of Fourier Transform

Significance of Time
scaling property



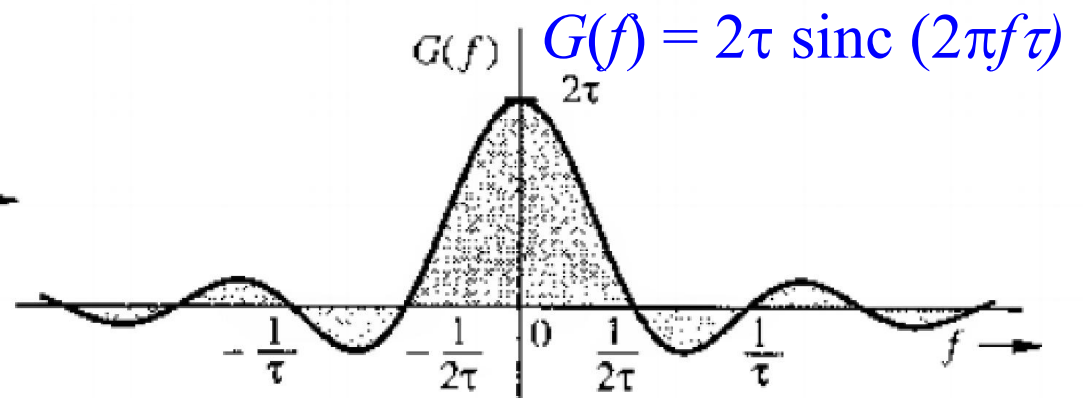
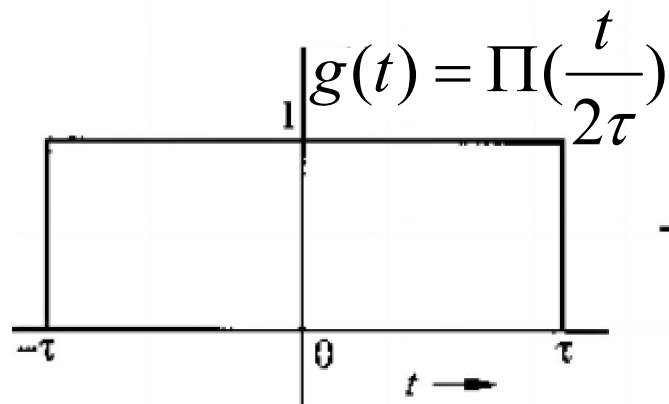
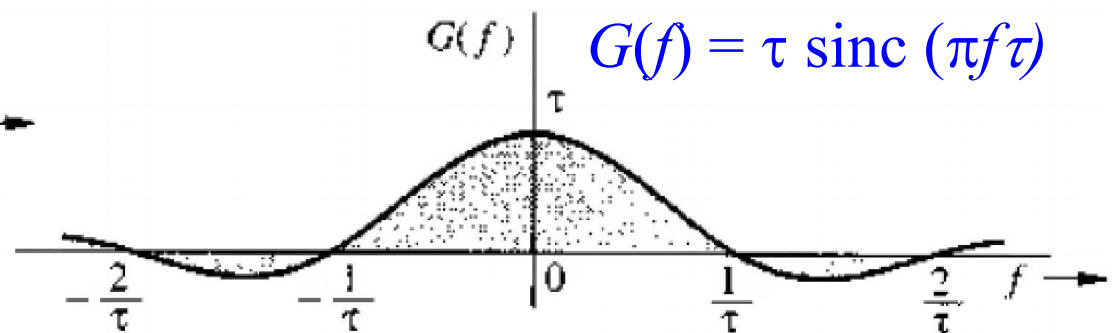
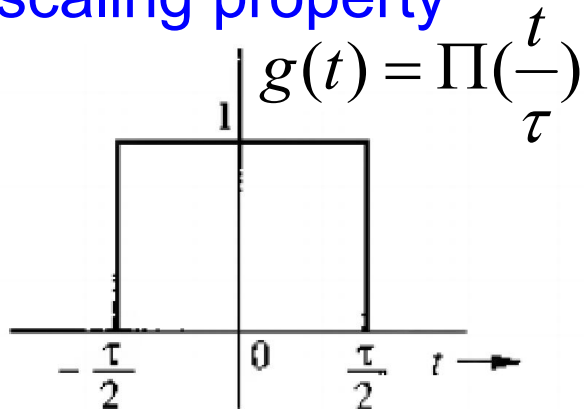
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Properties of Fourier Transform

Significance of Time scaling property

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Properties of Fourier Transform

Reflection property $g(-t) \Leftrightarrow G(-f)$

Directly results assuming $a = -1$ in scaling property

$$g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$$